Next-Day Rainfall Prediction Using Machine Learning

Antony Souza Siqueira
Federal University of Itajubá (UNIFEI)
Itajubá, Brazil

Augusto Juiz Ribeiro
Federal University of Itajubá (UNIFEI)
Itajubá, Brazil

Gustavo Totti Custódio dos Santos
Federal University of Itajubá (UNIFEI)
Itajubá, Brazil

Vitor Rocha Mininel
Federal University of Itajubá (UNIFEI)
Itajubá, Brazil

***Abstract*—This paper presents a machine learning pipeline for predicting next-day rainfall from daily meteorological observations. The problem is formulated as a binary supervised classification task using the *Rain in Australia* (weatherAUS) dataset, publicly available on Kaggle. The proposed workflow includes exploratory data analysis, missing value handling, categorical encoding, numerical scaling, stratified cross-validation, hyperparameter tuning, and comparison among different classifiers. Model evaluation prioritizes metrics suitable for imbalanced classification, such as recall, F1-score, and ROC-AUC. Among the evaluated models, Random Forest achieved the best overall performance, with an F1-score of 0.6589 and ROC-AUC of 0.8868 on the test set.**

***Index Terms*—machine learning, supervised classification, rainfall prediction, meteorological data, Random Forest**

I. Introduction

Rainfall prediction is a relevant practical task because it supports decision-making in areas such as agriculture, civil defense, transportation, logistics, tourism, and urban planning. The early identification of rainy days can help plan irrigation and harvesting activities, reorganize outdoor operations, reduce transportation risks, and prepare preventive responses to adverse weather events. In this context, even short-term forecasts, such as predicting whether it will rain on the next day, may contribute to safer and more efficient decisions.

Traditional weather forecasting is usually based on physical and numerical atmospheric models, which require specialized knowledge, robust computational infrastructure, and large volumes of observational data. Machine learning does not replace these models, but it can act as a complementary approach by exploring historical patterns in meteorological datasets. Supervised learning techniques are able to learn relationships among variables such as temperature, humidity, atmospheric pressure, wind, and precipitation, producing models that estimate the probability of future rainfall events [3], [5], [13].

In this work, rainfall prediction is formulated as a binary supervised classification problem. The model input is composed of meteorological attributes observed on the current day, and the output is the variable *RainTomorrow*, which indicates whether it will rain on the following day. The dataset used in this study is *Rain in Australia* (weatherAUS), made publicly available on Kaggle, containing approximately ten years of daily observations collected from several locations in Australia [1]. The variables in the dataset are consistent with historical meteorological records made available by the Australian Bureau of Meteorology, including temperature, humidity, pressure, wind, cloud cover, precipitation, and measurements taken at 9 a.m. and 3 p.m. [2].

The problem presents common challenges found in real-world data mining applications. The dataset contains missing values in different proportions, numerical and categorical variables, and class imbalance: the "no rain" class is considerably more frequent than the "rain" class. For this reason, accuracy alone may lead to misleading conclusions, since a trivial classifier that always predicts the majority class could obtain apparently high performance. Therefore, metrics such as recall, precision, F1-score, and ROC-AUC are required to better evaluate model behavior, especially regarding the ability to correctly identify rainy days [11], [12].

The objective of this work is to develop and evaluate a complete machine learning pipeline for next-day rainfall prediction. To achieve this objective, the study applies exploratory data analysis, structural cleaning, preprocessing, cross-validation, hyperparameter tuning, and comparison among different classification algorithms. The research question guiding the study is: how do different supervised models perform when predicting the minority class "rain tomorrow", considering tabular meteorological data and class imbalance? As a distinguishing aspect, the study does not rely only on accuracy: it considers evaluation metrics appropriate for imbalanced data and analyzes the meteorological relevance of the most important variables. The remainder of this paper is organized as follows: Section II presents the literature review, Section III describes the methodology, Section IV presents the experiments and results, Section V discusses the findings, and Section VI presents the conclusions and future work.

II. Literature Review

A. Supervised Learning and Binary Classification

Supervised learning consists of training models from labeled examples, that is, pairs formed by an input feature vector and a known output. In classification problems, the output belongs to a finite set of classes; in binary classification, there are only two possible outcomes, commonly represented by 0 and 1 [4], [5]. In this work, the positive class represents the occurrence of rainfall on the following day, while the negative class represents its absence.

In real-world applications, classifier performance depends not only on the selected algorithm but also on the quality of the data preparation steps. Missing values, different numerical scales, categorical attributes, and class imbalance can directly affect the learning process. For this reason, libraries such as scikit-learn are widely used because they provide integrated tools for preprocessing, pipeline construction, cross-validation, and model evaluation [13].

B. Machine Learning Applied to Meteorological Data

Data-driven methods have been explored in meteorology as alternatives or complements to traditional models. Navone and Ceccatto [14], for example, investigated the use of neural networks for monsoon rainfall prediction, showing the potential of computational techniques to capture complex climate patterns from historical data. More recently, deep learning methods have also been applied to atmospheric and climate problems, especially in tasks involving large volumes of data and nonlinear relationships [15].

Although physical atmospheric models remain essential for operational weather forecasting, machine learning techniques are useful in specific and well-defined tasks, such as rainfall occurrence classification, local pattern detection, and decision support. The approach adopted in this work is deliberately simpler and more interpretable: predicting next-day rainfall from tabular meteorological observations, without explicitly modeling the full atmospheric dynamics.

C. Related Works and Research Gap

Several studies have investigated the use of machine learning for rainfall and precipitation prediction. Sarasa-Cabezuelo [16] used the *Rain in Australia* dataset to compare algorithms such as KNN, Decision Tree, Random Forest, and neural networks, observing strong performance from nonlinear models. Raval *et al.* [17] also explored Australian rainfall data, comparing traditional models and neural networks in a workflow that included exploratory analysis, feature selection, and validation. Feng *et al.* [18] addressed probabilistic seasonal rainfall forecasting in Australia using Random Forest and large-scale climate indices, showing that machine learning models may outperform traditional statistical approaches in specific scenarios. From another perspective, Bagirov, Mahmood, and Barton [19] proposed a clusterwise linear regression approach for monthly rainfall prediction in Victoria, while Sachindra *et al.* [20] investigated machine learning techniques for statistical downscaling of precipitation.

These studies indicate that rainfall prediction is a recurrent and relevant problem, but they also show that no single technique is superior in all scenarios. Performance depends on the temporal scale, region, available variables, missing data strategy, and evaluation metrics. The gap addressed in this paper is the construction of an educational, reproducible, and comparative pipeline focused specifically on next-day rainfall classification, with emphasis on imbalanced data, stratified cross-validation, and interpretation of the most relevant meteorological variables.

D. Classification Models

Logistic Regression is one of the classical models for binary classification. Introduced in the context of binary sequence analysis by Cox [6], it estimates the probability of the positive class using the logistic function. Its main advantage is interpretability: the coefficients indicate the approximate direction and magnitude of the association between each variable and the probability of the positive class.

Decision Trees build classification rules through successive partitions of the feature space, using criteria such as Gini impurity or entropy. This type of model is intuitive and interpretable, but it may present high variance, especially when it grows without appropriate restrictions [7]. Random Forest, proposed by Breiman [8], reduces this instability by combining multiple trees trained on bootstrap samples of the data and using random subsets of features at each split. This strategy tends to improve generalization and capture nonlinear relationships among variables, which is useful in meteorological data.

Support Vector Machines (SVMs) were proposed by Cortes and Vapnik [9] as classifiers based on maximizing the margin between classes. Although SVMs with nonlinear kernels can capture complex decision boundaries, their computational cost may be high for large datasets. Therefore, linear implementations are viable alternatives when there are many observations or many variables after categorical encoding.

E. Validation, Metrics, and Imbalanced Data

Cross-validation is a fundamental technique for estimating the generalization ability of models. Instead of evaluating performance on a single data split, the training set is divided into k parts, alternating the portion used for validation. Kohavi [10] discusses cross-validation as an effective strategy for performance estimation and model selection.

In imbalanced classification problems, accuracy may be insufficient. If the majority class represents a large portion of the dataset, a classifier that always predicts this class may obtain high accuracy without properly detecting the minority class. He and Garcia [12] highlight that learning from imbalanced data requires attention both to model training and model evaluation. In this work, the positive class, corresponding to next-day rainfall, is the minority class. Therefore, metrics such as precision, recall, F1-score, and ROC-AUC are essential for model comparison [11].

III. Methodology

A. Dataset Description

The experiments were conducted using the *Rain in Australia* (`weatherAUS`) dataset, publicly available on Kaggle (*jsphyg/weather-dataset-rattle-package*). The dataset contains approximately 142,193 daily weather observations collected from 49 Australian weather stations over several years.

The prediction task is formulated as a binary classification problem in which the target variable, `RainTomorrow`, indicates whether rainfall occurs on the following day. The categorical labels were converted into binary values (`Yes` $\rightarrow$ 1 and `No` $\rightarrow$ 0).

The dataset is highly imbalanced, containing 110,316 observations (77.6%) belonging to the negative class and 31,877 observations (22.4%) belonging to the positive class. Consequently, evaluation metrics beyond accuracy were required to properly assess model performance.

A complete data dictionary, including variable descriptions, data types, missing value percentages, and meteorological interpretations, was generated and exported to `outputs/tables/data_dictionary.csv`.

B. Preprocessing

The preprocessing pipeline was implemented using the `Pipeline` and `ColumnTransformer` classes from scikit-learn. This design ensures that every preprocessing step is fitted exclusively on the training set, completely avoiding data leakage.

Before splitting the dataset, several structural cleaning operations were performed:

- Removal of 3,267 observations with missing values in the target variable (`RainTomorrow`);
- Conversion of `RainToday` and `RainTomorrow` from categorical values (`Yes`/`No`) to binary values (1/0);
- Extraction of two temporal attributes, `Month` and `Season`, from the original `Date` column, followed by removal of the original date attribute;
- Removal of variables with more than 40% missing values: `Evaporation` (43.2%), `Sunshine` (48.0%), and `Cloud3pm` (40.8%).

Inside the preprocessing pipeline, numerical variables were transformed using a `SimpleImputer(strategy='median')` followed by `StandardScaler`. Categorical variables were processed using `SimpleImputer(strategy='most_frequent')` followed by `OneHotEncoder(handle_unknown='ignore')`.

C. Train/Test Split and Cross-Validation

After preprocessing, the dataset was divided into training (80%) and testing (20%) subsets using `train_test_split` with stratification according to the target class and `random_state = 42`. The final dataset consisted of 113,754 training samples and 28,439 testing samples.

Model selection was performed using five-fold stratified cross-validation (`StratifiedKFold(n_splits=5, shuffle=True, random_state=42)`), preserving the original class distribution in every fold.

D. Evaluated Models

Five supervised classification algorithms were implemented and compared:

- **DummyClassifier**: `strategy='most_frequent'`; used as the baseline model.
- **Logistic Regression**: `class_weight='balanced'`, `max_iter=1000`.
- **Decision Tree**: `class_weight='balanced'`, optimized using GridSearchCV.
- **Random Forest**: `class_weight='balanced'`, `n_estimators=100`, optimized using RandomizedSearchCV.
- **Linear SVM**: `class_weight='balanced'`, `max_iter=2000`, calibrated using `CalibratedClassifierCV`. Due to computational constraints, this model was trained on a stratified sample of 30,000 observations from the training set.

E. Hyperparameter Optimization

Hyperparameter optimization was performed independently for the main predictive models.

For the Decision Tree classifier, GridSearchCV explored combinations of `max_depth`, `min_samples_split`, `min_samples_leaf`, and the splitting criterion (`gini` or `entropy`).

Random Forest optimization employed RandomizedSearchCV (`n_iter=20`), searching different values of `n_estimators`, `max_depth`, `min_samples_split`, and `min_samples_leaf`.

For Logistic Regression, GridSearchCV optimized the regularization parameter C over the values $\{0.01, 0.1, 1.0, 10.0\}$.

F. Evaluation Metrics

Model performance was evaluated using Accuracy, Precision, Recall, F1-score, ROC-AUC, and Average Precision (AP).

Since the dataset is considerably imbalanced, the F1-score was adopted as the primary model selection criterion, with ROC-AUC used as a secondary criterion in case of similar F1 values. Recall was also considered particularly important because missing an actual rainfall event has greater practical consequences than generating a false alarm.

IV. Experiments and Results

A. Dataset Characteristics

After structural preprocessing, the dataset contained 138,926 observations distributed over the remaining meteorological variables and engineered features. The target variable remained considerably imbalanced, with 110,316 observations (77.6%) belonging to class `No` and 31,877 observations (22.4%) belonging to class `Yes`.

The percentage of missing values varied substantially across variables. Three variables exceeded the predefined threshold of 40% missing values and were removed during preprocessing: `Evaporation` (43.2%), `Sunshine` (48.0%), and `Cloud3pm` (40.8%).

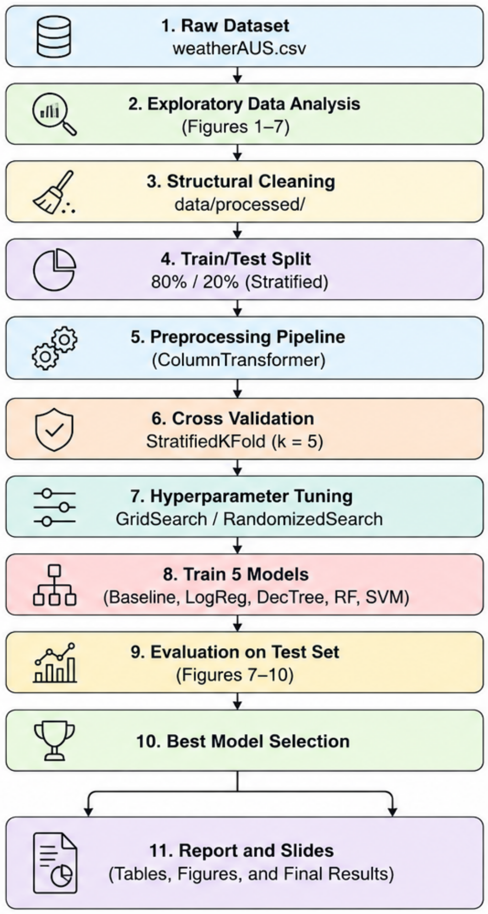

Figure 1. Overview of the proposed machine learning pipeline.

Pearson's correlation analysis identified `Humidity3pm` as the variable most positively correlated with rainfall occurrence ($r = 0.4462$), while `Pressure3pm` presented the strongest negative correlation ($r = -0.2260$). These findings are consistent with well-established meteorological principles.

B. Final Evaluation on the Test Set

Table 1 presents the final performance obtained on the independent test set.

Table I
PERFORMANCE METRICS ON THE TEST SET.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Baseline	0.7758	0.0000	0.0000	0.0000	0.5000
LogReg	0.7905	0.5221	0.7724	0.6230	0.8667
DecTree	0.7699	0.4914	0.7587	0.5965	0.8472
RF	**0.8483**	0.6643	0.6536	**0.6589**	**0.8868**
SVM	0.8442	**0.7258**	0.4904	0.5853	0.8647

The Random Forest classifier achieved the highest overall performance, obtaining the best F1-score and ROC-AUC while maintaining a balanced trade-off between Precision and Recall. Logistic Regression achieved the highest Recall, whereas Linear SVM obtained the highest Precision. The DummyClassifier confirmed that Accuracy alone is not an appropriate evaluation metric for this imbalanced dataset, achieving an apparently high Accuracy (77.58%) while presenting an F1-score of zero.

The confusion matrices, ROC curves, and Precision–Recall curves further confirm the superior discriminative capability of the Random Forest model. Additionally, the feature importance analysis identified `Humidity3pm` and `Pressure3pm` as the most influential variables, reinforcing the conclusions obtained from the exploratory data analysis and demonstrating consistency between statistical analysis and machine learning interpretation.

V. Discussion

A. Comparative Performance

The Random Forest model achieved the best overall performance, obtaining an F1-score of 0.6589 and a ROC-AUC of 0.8868 on the test set, outperforming the baseline by approximately 65.9 percentage points in terms of F1-score. This superior performance can be explained by the ensemble learning strategy adopted by Random Forest, which combines multiple decision trees trained on bootstrap samples with random subsets of features. This approach reduces model variance, improves generalization, and enables the capture of complex non-linear relationships between meteorological variables and rainfall occurrence.

Although Logistic Regression is a simple linear model, it achieved a competitive F1-score of 0.6230, demonstrating that linear relationships between meteorological variables and the occurrence of rainfall provide substantial predictive information. Furthermore, it obtained the highest Recall (0.7724) among all evaluated models, making it particularly suitable for applications where minimizing false negatives is a priority.

The Decision Tree classifier, after hyperparameter tuning, obtained an F1-score of 0.5965, which was lower than that achieved by Logistic Regression. Although decision trees are highly interpretable, they are more susceptible to overfitting and exhibit higher variance than ensemble methods, explaining their inferior predictive performance compared to Random Forest.

The Linear SVM achieved the highest Precision (0.7258), indicating that when it predicts rainfall, it is usually correct. However, its relatively low Recall (0.4904) indicates that it failed to identify a considerable proportion of rainy days, making it less suitable for scenarios where detecting as many rainfall events as possible is essential.

B. Precision–Recall Trade-off

The use of `class_weight='balanced'` increased the sensitivity of the models toward the minority class (*RainTomorrow* = *Yes*), resulting in higher Recall values. This behavior is appropriate for the rainfall prediction problem because failing to predict an actual rainfall event generally has greater practical consequences than issuing a false alarm.

The evaluated models exhibited different trade-offs between Precision and Recall. Logistic Regression prioritized Recall (0.7724) at the expense of Precision (0.5221), making it appropriate for early warning systems where missing a rainy day is costly. Conversely, the Linear SVM favored Precision (0.7258) but achieved a considerably lower Recall (0.4904), indicating a conservative prediction strategy. Random Forest provided the best balance between both metrics, with Precision of 0.6643 and Recall of 0.6536, resulting in the highest F1-score among all evaluated models.

C. Most Relevant Variables

The feature importance analysis revealed that `Humidity3pm` and `Pressure3pm` were the most influential predictors in the Random Forest model. High relative humidity during the afternoon indicates atmospheric conditions close to saturation, increasing the likelihood of precipitation. Conversely, low atmospheric pressure is commonly associated with approaching weather fronts and cyclonic systems that favor rainfall.

These findings are consistent with the correlation analysis performed during the exploratory data analysis, where `Humidity3pm` presented the strongest positive correlation with the target variable, while `Pressure3pm` exhibited the strongest negative correlation. Therefore, the model's interpretation agrees with well-established meteorological knowledge.

D. Limitations

Despite the satisfactory predictive performance, this study presents some limitations. First, the dataset contains observations exclusively from Australia; therefore, the trained models cannot be directly generalized to other climatic regions without retraining and validation.

Second, important variables such as `Sunshine` and `Evaporation` contained a high percentage of missing values and had to be removed during preprocessing, potentially reducing the predictive power of the models.

Third, the proposed pipeline assumes that observations are independent and identically distributed, ignoring the temporal dependence naturally present in weather data. Time-series approaches could potentially exploit this sequential structure to improve predictive performance.

Finally, due to computational constraints, the Linear SVM model was trained using a stratified sample of 30,000 observations instead of the complete training set, which may have limited its overall performance.

VI. Conclusion and Future Work

This work developed a complete and reproducible machine learning pipeline for next-day rainfall prediction using the Rain in Australia dataset. Five supervised classification algorithms were implemented, optimized, and compared using evaluation metrics appropriate for imbalanced datasets, including Precision, Recall, F1-score, ROC-AUC, and Average Precision.

Among the evaluated models, Random Forest achieved the best overall performance, obtaining an F1-score of 0.6589 and a ROC-AUC of 0.8868. The feature importance analysis identified `Humidity3pm` and `Pressure3pm` as the most relevant predictors, which is consistent with established meteorological knowledge regarding atmospheric humidity and pressure variations preceding rainfall events.

The entire pipeline is fully reproducible. By executing `python main.py` with the `weatherAUS.csv` dataset placed in the `data/raw/` directory, all preprocessing steps, model training, evaluation metrics, figures, tables, and the exported best model can be regenerated automatically.

Future work may explore more advanced machine learning techniques, including gradient boosting methods such as XGBoost, LightGBM, and CatBoost. Additionally, time-series models such as Long Short-Term Memory (LSTM) networks and Temporal Fusion Transformers could better exploit the temporal dependence between consecutive days. Other promising directions include evaluating models separately for different weather stations, calibrating predicted probabilities using Platt Scaling or Isotonic Regression, incorporating more recent meteorological data through the Australian Bureau of Meteorology API, and developing an interactive web application or REST API to facilitate practical deployment of the proposed solution.

References

[1] J. P. Young, "Rain in Australia," Kaggle, 2017. [Online]. Available: https://www.kaggle.com/datasets/jsphyg/weather-dataset-rattle-package. Accessed: Jun. 26, 2026.

[2] Bureau of Meteorology, Australian Government, "Climate Data Online." [Online]. Available: https://www.bom.gov.au/climate/data/. Accessed: Jun. 26, 2026.

[3] A. Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, 3rd ed. Sebastopol, CA, USA: O'Reilly Media, 2022.

[4] C. M. Bishop, *Pattern Recognition and Machine Learning*. New York, NY, USA: Springer, 2006.

[5] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, 2nd ed. New York, NY, USA: Springer, 2009.

[6] D. R. Cox, "The regression analysis of binary sequences," *Journal of the Royal Statistical Society: Series B*, vol. 20, no. 2, pp. 215–232, 1958, doi: 10.1111/j.2517-6161.1958.tb00292.x.

[7] L. Breiman, J. H. Friedman, R. A. Olshen, and C. J. Stone, *Classification and Regression Trees*. Belmont, CA, USA: Wadsworth, 1984.

[8] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001, doi: 10.1023/A:1010933404324.

[9] C. Cortes and V. Vapnik, "Support-vector networks," *Machine Learning*, vol. 20, no. 3, pp. 273–297, 1995, doi: 10.1007/BF00994018.

[10] R. Kohavi, "A study of cross-validation and bootstrap for accuracy estimation and model selection," in *Proceedings of the 14th International Joint Conference on Artificial Intelligence*, Montreal, Canada, 1995, pp. 1137–1145.

[11] T. Fawcett, "An introduction to ROC analysis," *Pattern Recognition Letters*, vol. 27, no. 8, pp. 861–874, 2006, doi: 10.1016/j.patrec.2005.10.010.

[12] H. He and E. A. Garcia, "Learning from imbalanced data," *IEEE Transactions on Knowledge and Data Engineering*, vol. 21, no. 9, pp. 1263–1284, 2009, doi: 10.1109/TKDE.2008.239.

[13] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[14] H. D. Navone and H. A. Ceccatto, "Predicting Indian monsoon rainfall: A neural network approach," *Climate Dynamics*, vol. 10, no. 6–7, pp. 305–312, 1994.

[15] S. Rasp, M. S. Pritchard, and P. Gentine, "Deep learning to represent subgrid processes in climate models," *Proceedings of the National Academy of Sciences*, vol. 115, no. 39, pp. 9684–9689, 2018, doi: 10.1073/pnas.1810286115.

[16] A. Sarasa-Cabezuelo, "Prediction of rainfall in Australia using machine learning," *Information*, vol. 13, no. 4, p. 163, 2022, doi: 10.3390/info13040163.

[17] M. Raval, P. Sivashanmugam, V. Pham, H. Gohel, A. Kaushik, and Y. Wan, "Automated predictive analytics tool for rainfall forecasting," *Scientific Reports*, vol. 11, article 17704, 2021, doi: 10.1038/s41598-021-95735-8.

[18] P. Feng, B. Wang, D. L. Liu, F. Ji, X. Niu, H. Ruan, L. Shi, and Q. Yu, "Machine learning-based integration of large-scale climate drivers can improve the forecast of seasonal rainfall probability in Australia," *Environmental Research Letters*, vol. 15, no. 8, 2020, doi: 10.1088/1748-9326/ab9e98.

[19] A. M. Bagirov, A. Mahmood, and A. Barton, "Prediction of monthly rainfall in Victoria, Australia: Clusterwise linear regression approach," *Atmospheric Research*, vol. 188, pp. 20–29, 2017, doi: 10.1016/j.atmosres.2017.01.003.

[20] D. A. Sachindra, K. Ahmed, M. M. Rashid, S. Shahid, and B. J. C. Perera, "Statistical downscaling of precipitation using machine learning techniques," *Atmospheric Research*, vol. 212, pp. 240–258, 2018, doi: 10.1016/j.atmosres.2018.05.022.